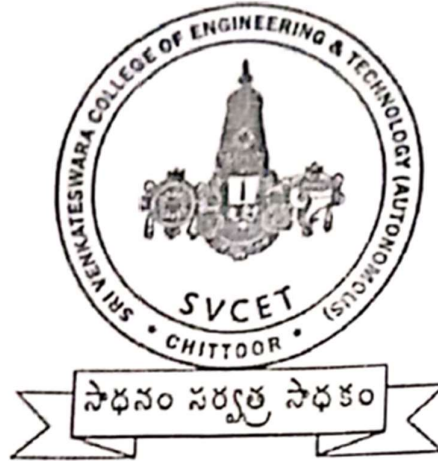


SRI VENKATESWARA COLLEGE OF ENGINEERING & TECHNOLOGY

(Autonomous)



NAME OF THE STUDENT :

ROLL NUMBER :

BRANCH & YEAR : CSD-C or CSM-C (ASS-2)

SUBJECT WITH CODE : BASIC ELECTRICAL & ELECTRONICS

TOPIC : Energy resources ,Electricity bill and
safety measures

SUBMITTED ON :

SUBMITTED TO :

1) Explain the wheatstone bridge?

2.9 Wheatstone bridge.

A very important device used in the measurement of medium resistances is the Wheatstone bridge. A Wheatstone bridge has been in use longer than almost any electrical measuring instrument. It is still an accurate and reliable instrument and is extensively used in industry.

The Wheatstone bridge is an instrument for making comparison measurements and operates upon a null indication principle. This means the indication is independent of the calibration of the null indicating instrument or any of its characteristics.

For this reason, very high degrees of accuracy can be achieved using Wheatstone bridge. Accuracy of 0.1% is quite common with a Wheatstone bridge as opposed to accuracies of 3% to 5% with ordinary ohmmeter **for measurement of medium resistances.**

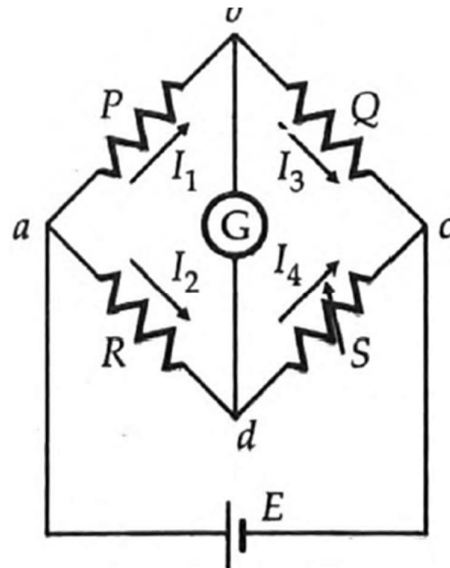


Fig. shows the basic circuit of a Wheatstone bridge. It has four resistive arms, consisting of resistances P, Q, R and S together with a source of emf (a battery) and a null detector, usually a galvanometer G or other sensitive current meter. The current through the galvanometer depends on the potential difference between points c and d.

The bridge is said to be balanced when there is no current through the galvanometer or when the potential difference across the galvanometer is zero. This occurs when the voltage from point **b** to point **a** equals the voltage from point **d** to point **b**; or, by referring to the other battery terminal, when the voltage from point **d** to point **c** equals the voltage from point **b** to point **c**.

For bridge balance, we can write, $I_1P = I_2R$ eq. (3.1)

For the galvanometer current to be zero, the following conditions also exist:

$$I_1 = I_3 = \frac{E}{P+Q} \quad \text{and} \quad I_2 = I_4 = \frac{E}{R+S} \quad \text{where } E = \text{emf of the battery.} \quad \text{eq. (3.2)}$$

Combining Eqns. 3.1 and 3.2 and simplifying, we obtain :

$$\frac{P}{P+Q} = \frac{R}{R+S} \quad \text{from which } \mathbf{QR = PS} \quad \text{eq. (3.3)}$$

eq 3.3 is the well known expression for the balance of Wheatstone bridge. If three of the resistances are known, the fourth may be determined from eq.3.3

$$\text{and we obtain : } R = S \frac{P}{Q}$$

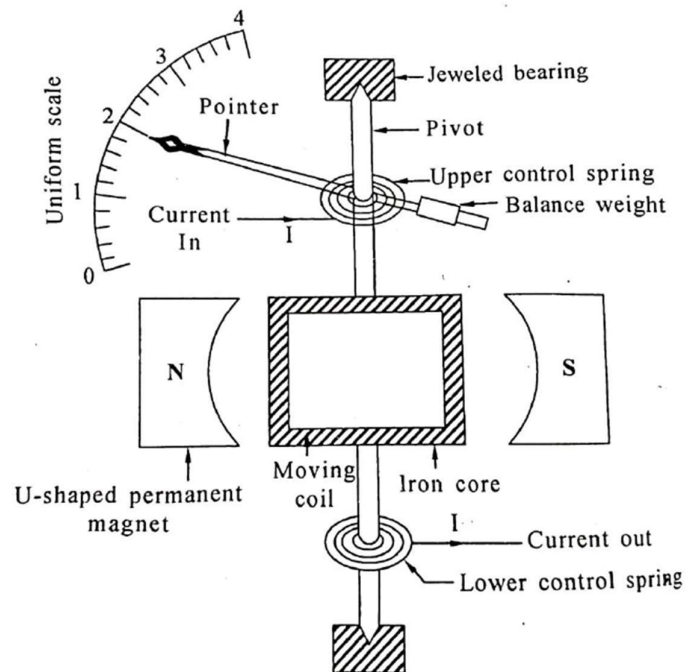
where R = unknown resistance,

S = '*standard arm*' of the bridge and

P and Q = '*ratio arms*'.

2) Explain the construction and working principle of PMMC and MI Instruments

2.7 Permanent Magnet Moving Coil (PMMC) instrument



Two spiral springs of phosphor bronze are mounted one above the other below the coil. The springs serve 2 purposes. It leads the current into and out of the coil and also provides the controlling torque. A light pointer is attached to the spindle which moves over a calibrated scale. The damping torque is provided by eddy currents induced in the aluminium former.

Working

- When the instrument is connected in the circuit to measure current and voltage, the operating current flows through the coil.
- When the current is passed through the coil, a deflecting torque is produced due to the reaction between permanent magnetic field and the magnetic field of the coil. This deflecting torque is proportional to the value of current in the coil

$$T_d \propto I.$$

- Due to this torque, the coil moves and the pointer deflects over the calibrated scale. When the coil moves, the springs are twisted or rewound, thus the spring offers a controlling torque to the moving system.
- The controlling torque due to spring is directly proportional to the deflection of the moving part $T_c \propto \theta$. When the deflecting and controlling torque are equal

$$T_d = T_c.$$

- The pointer comes to rest at its final position, hence, $\theta \propto I$. Since the deflection (θ) is directly proportional to the current (I), these instruments have uniform scale.

Deflecting torque: It is directly proportional to the current or the voltage to be measured. So, the instrument can be used to measure direct current and DC voltage.

Control torque : The torque which controls the deflecting torque is called control torque (Spring control). It brings the pointer back to zero position when the deflecting torque is removed.

Damping torque: Eddy current damping. When the moving coil made of aluminium former is moved due to the force exerted on it, it cuts the magnetic flux lines produced by the permanent magnet. Hence, eddy currents are induced in the former.

2.8 Moving Iron (MI) Instruments

In this type of instruments ammeters and voltmeters the current carrying coil is stationary but the iron core rotates hence it is called moving iron instruments. These instruments are mostly used to measure the current and voltage in an alternating and direct current circuits. Therefore there are 2 types of moving iron instruments

(i) Attractive type: These operate on the principle of attraction of a single piece of soft iron into a magnetic field

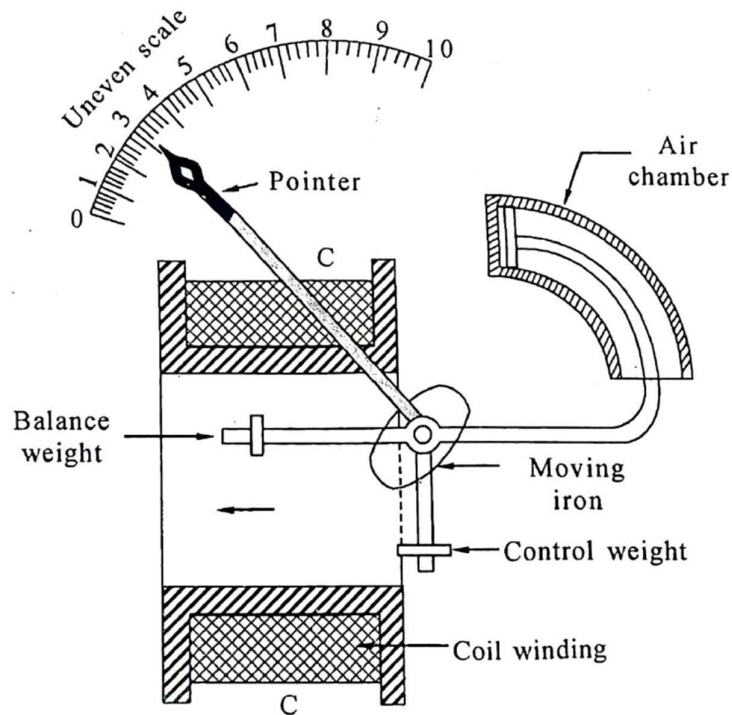
(ii) repulsion type: It operates on the principle of repulsion of 2 adjacent iron pieces magnetized by the same magnetic field

Attraction type moving iron instrument:

It works on the principle of attraction of a single piece of soft iron into a magnetic field.

Construction: Fig. shows the construction details of attraction type moving iron instrument. It consists of a fixed coil or solenoid through which the current is passed. An oval shaped soft iron is attached to the spindle in such a way that it can move in and out of the coil. A pointer is attached to the spindle, so that it may deflect along with the moving iron over a calibrated scale to stop the spindle is pivoted in jewel bearings at both the ends. A spiral spring is attached

to the spindle to provide controlling torque. An aluminum vane is attached to the spindle which moves in a closed air chamber to provide damping torque.



Working:

Working:

- when the instrument is connected in the circuit to measure current or voltage, the operating current flows through the coil.
- Whenever current flows through the coil the electromagnetic field is set up along its axis. In other words, the coil behaves like a magnet and therefore it attracts the soft iron piece towards it, thus providing the **deflecting torque** (T_d)
- The deflecting torque causes the pointer attached to the moving system moves from zero position over a calibrated scale. The deflecting torque depends upon the value of magnetic flux (current flowing through the coil).
- The spiral spring offers the controlling torque which depends upon the angle of twist. When the deflecting torque is equal to the controlling torque ($T_d = T_c$) the pointer will come to rest at a position.
- The deflecting torque (T_d) is proportional to square off the current I^2 and controlling torque (T_c) is proportional to angle of twist ' θ '. At a steady position of deflection, $T_d = T_c$, hence $\theta \propto I^2$. In case of AC measurement, $\theta \propto I_{rms}^2$.
- Hence, the scale is not uniform, being crowded at the beginning and spread out near the finishing end of the scale.

3) Explain single phase transformer, principle and operation

Working principle:

The basic principle of a transformer is **electromagnetic induction**. A 1 ϕ transformer consists of 2 windings placed over a laminated silicon steel core. The winding having less number of turns is called low-voltage winding and the winding having more number of turns is called high voltage winding.

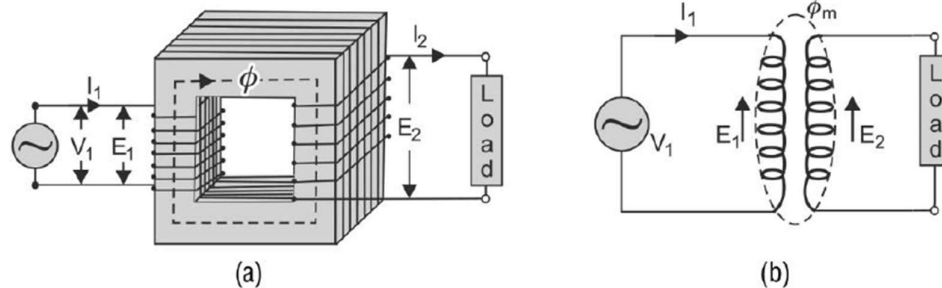


Fig. 2.2 (a) Single-phase transformer (core and windings)

(b) Flux linking with primary and secondary

- The winding to which AC supply is connected is called a **primary winding** and the other one is called a **secondary winding** to which load is connected.
- Once AC supply of voltage V_1 is given to primary winding, an alternating flux is set-up in the magnetic core which links with the primary and secondary winding. Consequently, self-induced emf E_1 and mutually-induced emf E_2 are induced in primary and secondary, respectively.
- These induced emf's are developed in phase opposition to V_1 as per Lenz's law. The self-induced emf in the primary is also called back emf since it acts in opposite direction to the applied voltage.
- Although, there is no electrical connection between primary and secondary winding, still electric power is transferred from one circuit (primary side) to the other circuit (secondary side). It is because of **magnetic coupling**, i.e., the alternating flux which is set-up in the core linking with both the windings.
- The magnitude of induced emf in a coil depends upon rate of change of flux linkages i.e., $e \propto N$. since, the rate of change of flux for both the winding is the same, the magnitude of induced emf in primary and secondary will depend upon their number of turns, i.e., primary induced emf $E_1 \propto N_1$ and secondary induced emf $E_2 \propto N_2$.
- When $N_2 > N_1$, the transformer is called a **step-up transformer**, when $N_2 < N_1$ the transformer is called **step-down transformer**.

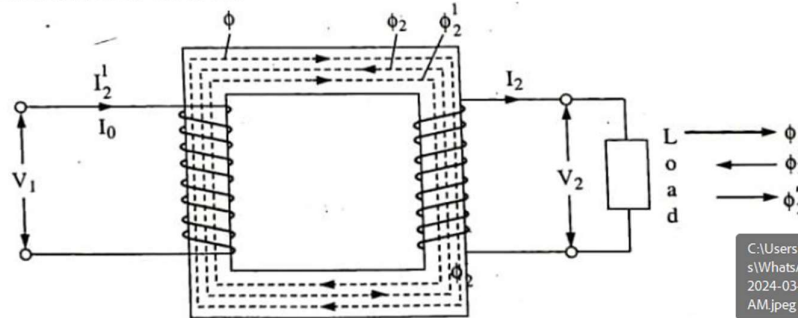
Turn ratio: The ratio of primary to secondary turns is called turn ratio, i.e.,

$$\text{turn ratio} = \frac{N_1}{N_2}$$

Transformation ratio: The ratio of secondary voltage to primary voltage is called voltage transformation ratio of the transformer. It is represented by K.

$$K = \frac{V_2}{V_1} = \frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{I_1}{I_2}$$

Transformer on load:



Construction: The main parts of a transformer are:

- (i) **Core:** The purpose of core is to provide a path of low reluctance for magnetic flux.
- (ii) **Windings:** These are made up of copper which are placed on transformer core. The windings connected to supply is known as primary winding and the winding connected to load is called secondary winding.
- (iii) **Tank:** The transformer with core and windings is placed in proper container called transformer tank.
- (iv) **Transformer oil:** The insulating oil which is used in the tank of transformer is called transformer oil. It provides 3 functions.
 - It carries away the heat produced in the core and windings.
 - Additional insulation for windings.
 - Protects the insulation from dust and moisture.
- (v) **Conservator:** A conservator performs the following functions: It maintain the oil level in the tank. It provides space for expansion of oil when temperature of transformer increases.
- (vi) **Breather:** The function of breather is to prevent entry of moisture in the air and allows dry air into the transformers. Moisture reduces the dielectric strength of the oil.
- (vii) **Terminal bushings:** Connections from the transformer windings are brought out by means of bushings mounted on the transformer tank.
- (viii) **Explosion vent:** It provides instantaneous relieving of dangerous pressure and protects the transformer tank against explosion.
- (ix) **Oil gauge:** For indicating the level of the oil in the conservator, an oil level indicator is fixed to eat. It is essential that the oil level in the conservator is maintained above the predetermined minimum level.
- (x) **Buchholz relay:** It is used for protection of oil filled transformers from incipient faults below oil level.

4) explain the induction motor and alternator principle and construction?

Ans :A Induction motor

The Induction machines are also called asynchronous machines i.e., the machines which never run at a synchronous speed. Whenever we say induction machine we mean to say induction motor. Induction motors may be single-phase or three-phase. The single phase induction motors are usually built in small sizes (upto 3 H.P). Three phase induction motors are the most commonly used AC motors in the industry because they have simple and rugged construction, low cost, high efficiency, reasonably good power factor, self-starting and low maintenance cost. Almost more than 90% of the mechanical power used in industry is provided by three phase induction motors.

Working principle:

When 3-phase supply is given to the stator winding of a 3-phase wound induction motor, a revolving field is set up in the stator core. The resultant magnetic field set-up by the stator core, at any instant, is shown in Fig. The direction of the resultant field is marked by an arrow head F_m . As per the supply sequence, let this field is rotating in an anti-clockwise direction at synchronous speed ω_s radian per second.

The revolving field is cut by the stationary rotor conductors and an emf is induced in the rotor conductors. Since the rotor conductors are short circuited, current flows through them in the direction as marked in Fig. 9.9(a). A resultant field F_r is set-up by the rotor current carrying conductors. This field tries to come in line with the stator revolving field F_m , due to which an electromagnetic torque T_e develops and rotor starts rotating in same direction as that of stator revolving field.

Construction

A 3-phase induction motor consists of two main parts, namely stator and rotor.

1. Stator: It is the stationary part of the motor. It has three main parts, namely. (i) Outer frame, (ii) Stator core and (iii) Stator winding.

(i) Outer frame: It is the outer body of the motor. Its function is to support the stator core and to protect the inner parts of the machine. For small machines the frame is casted but for large machines it is fabricated. To place the motor on the foundation, feet are provided in the outer frame as shown in Fig

(ii) Stator core: When AC supply is given to the induction motor, an alternating flux is set up in the stator core. This alternating field produces hysteresis and eddy current loss. To minimise these losses, the core is made of high grade silicon steel stampings.

The laminations are assembled under hydraulic pressure and are keyed to the frame. Each stamping is insulated from the other with a thin varnish layer. Slots are punched on the inner periphery of the stampings, as shown in Fig., to accommodate stator winding.

(iii) Stator winding: The stator core carries a three phase winding which is usually supplied from a three phase supply system. The six terminals of the winding (two of each phase) are connected in the terminal box of the machine. The stator of the motor is wound for definite number of poles, the exact number being determined by the requirement of speed. It will be seen that greater the number of poles, the lower is the speed and vice-versa, since

$$N_s \propto \frac{1}{P} \quad \left(\because N_s = \frac{120f}{P} \right)$$

$\therefore$. The three- phase winding may be connected in star or delta externally through a starter.

2. Rotor: The rotating part of the motor is called rotor. 2 types of rotors are used for 3-phase induction motors. (i) Squirrel cage rotor (ii) Phase wound rotor.

(i) Squirrel cage rotor: A squirrel cage rotor consists of a laminated cylindrical core having semi-closed circular slots at the outer periphery. Copper or aluminium bar conductors are placed in these slots and short circuited at each end by copper or aluminium rings, called short circuiting rings, as shown in Fig. In this, the rotor winding is permanently short-circuited and no external resistance can be added in the rotor circuit. Fig. clearly shows that the slots are not parallel to the shaft but these are skewed. The skewing provides the following advantages:

- (a) Humming is reduced, that ensures quiet running.
- (b) At different positions of the rotor, smooth and sufficient torque is obtained.
- (c) It reduces the magnetic locking of the stator and rotor,
- (d) It increases the rotor resistance due to the increased length of the rotor bar conductors.

(ii) Slip ring rotor: This rotor is also cylindrical in shape which consists of large number of stampings. A number of semi-closed slots are punched at its

outer periphery. A 3-phase insulated winding is placed in these slots. The rotor is wound for the same number of poles as that of stator. The rotor winding is connected in star and its remaining three terminals are connected to the slip rings. The rotor core is keyed to the shaft. Similarly, slip-rings are also keyed to the shaft but these are insulated from the shaft. (see Fig.)

In this case, depending upon the requirement any external resistance can be added in the rotor circuit. In this case also the rotor is skewed. A mild steel shaft is passed through the centre of the rotor and is fixed to it with key. The purpose of shaft is to transfer mechanical power.

Slip:

In an induction motor, the speed of rotor is always less than synchronous speed. The difference between the speed of revolving field (N_s) and the rotor speed (N) is called slip.

The slip is usually expressed as a percentage of synchronous speed (N_s) and is represented by symbol S .

Mathematically, % Slip,
$$\%S = \frac{N_s - N}{N_s} \times 100$$

$$S = \frac{N_s - N}{N_s}$$

Rotor speed,
$$N = N_s (1 - S)$$

The difference between synchronous speed and rotor speed is called slip speed
i.e., Slip speed = $N_s - N$

B Alternator :

Alternator

The mechanical power or energy is converted into electrical power or energy with the help of an AC machine called alternator or synchronous generator. However, when the same machine can be used to convert electrical power or energy into mechanical power or energy, then it is known as a synchronous motor.

Thus, the same machine can be operated as a generator or as a motor and in general, it is called as a **synchronous machine**. In fact, it is a machine which rotates only at synchronous speed ($N_s = 120 f/P$) under all conditions. To understand the construction, working and performance of the machine its study is divided into number of main topics as chapters, to be followed.

Working principle

When a conductor or coil cuts across the magnetic field an emf is induced in it by the phenomenon called electromagnetic induction. This can be achieved either by rotating a coil in the stationary magnetic field or by keeping the coil stationary and rotating the magnetic field. (The magnetic field can be rotated by placing the field winding on the rotating part of the machine).

Construction:

The important parts of a synchronous machine are (1) Stator (2) Rotor

1. Stator: The outer stationary part of the machine is called stator; it has the following important parts:

(i) Stator frame: It is the outer body of the machine made of cast iron and it protects the inner parts of the machine. It can be also made of any other strong

material since it is not to carry the magnetic field. Cast iron is used only because of its high mechanical strength.

(ii) Stator Core: The stator core is made of silicon steel material. It is made from number of stamping which are insulated from each other. Its function is to provide an easy path for the magnetic lines of force and accommodate the stator winding.

(iii) Stator Winding: Slots are cut on the inner periphery of the stator core in which three-phase or one-phase winding is placed. Enamelled copper is used as winding material.

2. Rotor: The rotating part of the machine is called rotor. From construction point of view, there are two types of rotors named as

(i) Salient pole type rotor; (ii) Non-salient pole type rotor.

(i) Salient pole type rotor: In this case, projected poles are provided on the rotor. The cost of construction of salient pole type rotors is low, moreover sufficient space is available to accommodate field winding but these cannot bear high mechanical stresses at high speeds. Therefore, salient pole type construction is suited for medium and low speeds and are usually employed at hydro-electric and diesel power plants as synchronous generators. Since the speed of these machines (generators) is quite low, to obtain the required frequency, the machines have large number of poles as shown in Figs. 6.7 and 6.8. To accommodate such a large number of poles, these machines have larger diameter and small length.

5) Explain dc generator and dc motor principle and construction

Ans :

A: Dc generator :

DC Generator

Working principle:

- A DC generator is a dynamic DC machine which generates electrical energy from mechanical energy. The emf induced is called dynamically induced emf.
- It operates on the principle of **Faraday's law of electromagnetic induction**, which states that whenever a current carrying conductor cuts the magnetic flux, a dynamically induced emf is generated.
- Its magnitude depends on the rate of change of magnetic flux linked with the conductor. If the conductor is connected to a closed circuit, then a current will flow through it. The basic elements required in a DC generator to generate a dynamically induced emf are:
 - A steady magnetic field
 - A current carrying conductor and
 - Relative motion between conductor and magnetic field

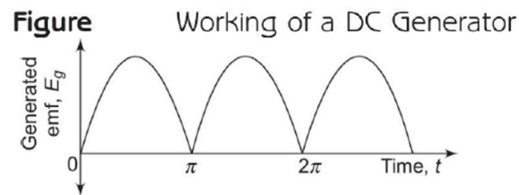
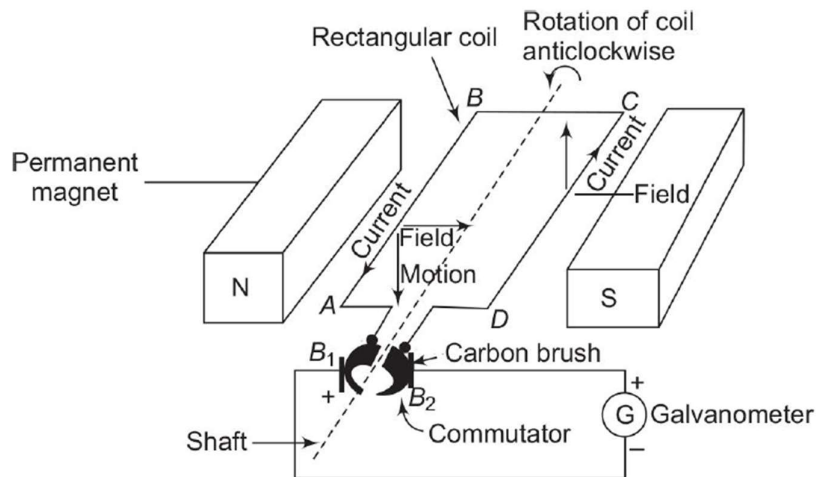


Fig Unidirectional emf Generated in a DC Generator

- The basic working of a DC generator can be explained with the help of a single rectangular coil ABCD, as shown in Fig. This rectangular coil is placed between two opposite poles of the magnet.
- When the field coil wound over the magnet is excited by the DC source, the magnets get energised and magnetic flux is generated between these opposite poles. The rectangular coil connected to the shaft gets rotated in a specific direction driven by the prime mover.
- Therefore, the rectangular coil, when it rotates about its axis in the magnetic field i.e., when it moves from horizontal to vertical position, it cuts the magnetic field generated by the field windings. Hence, an emf is induced in both the sides of the coil i.e., on both AB and CD.
- Since the loop is closed, a current will be circulated through the loop. Using Fleming's right-hand rule, the direction of current can be determined. The emf generated in a DC generator is shown in Fig.

Construction:

The schematic diagram of a DC generator with necessary parts is shown in Fig.

The main components are:

- (i) Magnetic frame (or) yoke
- (ii) Pole cores and pole shoes
- (iii) Field or exciting windings
- (iv) Armature core and windings
- (v) Armature windings
- (vi) Commutators
- (vii) Brushes
- (viii) Bearings
- (ix) Shaft

The above 9 parts can be grouped into 4 major components, namely magnetic field system, armature, commutator and brush assembly.

Stator magnetic field system

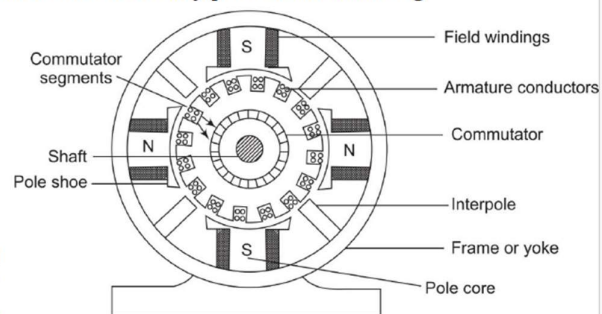


Fig Schematic Diagram of a DC Generator

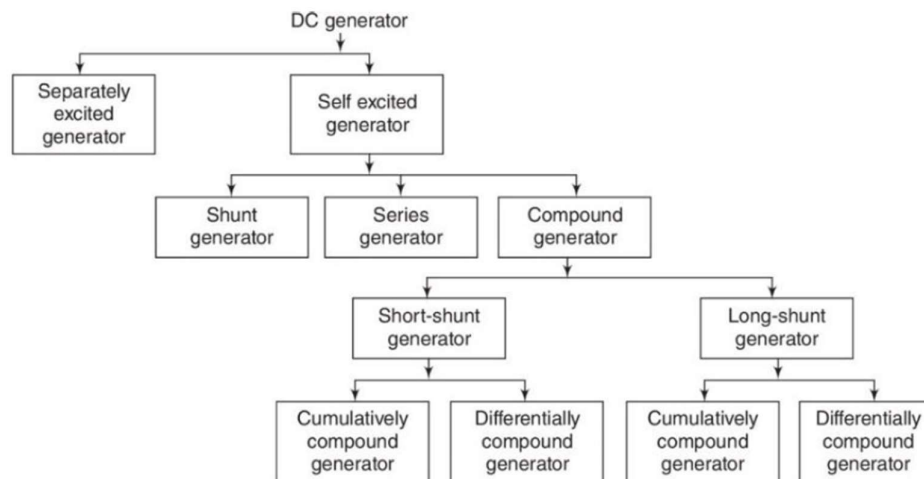


Fig Types of DC generator

DC motor :

The machine when used to convert DC electrical power into mechanical power, it known as a **DC motor**. From construction point of view there is no difference between a DC generator and motor. The DC motors are very useful where wide range of speeds and perfect speed regulation is required such as electric traction.

An electro-mechanical energy conversion device (electrical machine) that converts DC electrical energy or power (EI) into mechanical energy or power (Z T) is called a DC motor. Electric motors are used for driving industrial machines, e.g., hammers, presses, drilling machines, lathes, rollers in paper and steel industry, blowers for furnaces, etc., and domestic appliances, e.g., refrigerators, fans, water pumps, toys, mixers, etc. The block diagram of energy conversion, when the electro-mechanical device works as a motor, is shown in Fig. 5.1.

Construction

The construction of a DC motor is exactly **similar to DC generator**. The important parts of a DC motor are: yoke or frame, main field system, brushes, armatures and commutator. The functions of certain components vary with respect to a DC motor. In a DC motor, the commutator is used to convert the alternating torque produced in the armature into a unidirectional torque. A separate supply is given to armature winding to produce the required torque in the armature. The electrical power is converted into a mechanical power in the armature winding.

Working Principle:

The operation of a DC motor is based on the principle that when a current carrying conductor is placed in a magnetic field, a mechanical force is experienced by it. The direction of this force is determined by Fleming's Left Hand Rule and its magnitude is given by the relation:

$$F = Bil \text{ newton}$$

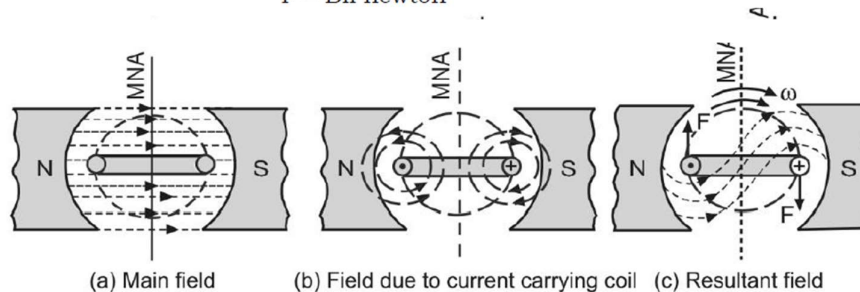


Fig. Working principle of a motor

For simplicity, consider only one coil of the armature placed in the magnetic field produced by a bipolar machine [see Fig.(a)]. When DC supply is connected to the coil, current flows through it which sets up its own field as shown in Fig.(b). By the interaction of the two fields (i.e., field produced by the main poles and the coil), a resultant field is set up as shown in Fig.(c). The tendency of this is to come to its original position i.e., in straight line due to which force is exerted on the two coil sides and torque develops which rotates the coil.

Alternately, it can be said that the main poles produce a field F_m . Its direction is marked in Fig. When current is supplied to the coil (armature conductors), it produces its own field marked as F_r . This field tries to come in line with the main field and an electromagnetic torque develops in clockwise direction as marked in Fig.